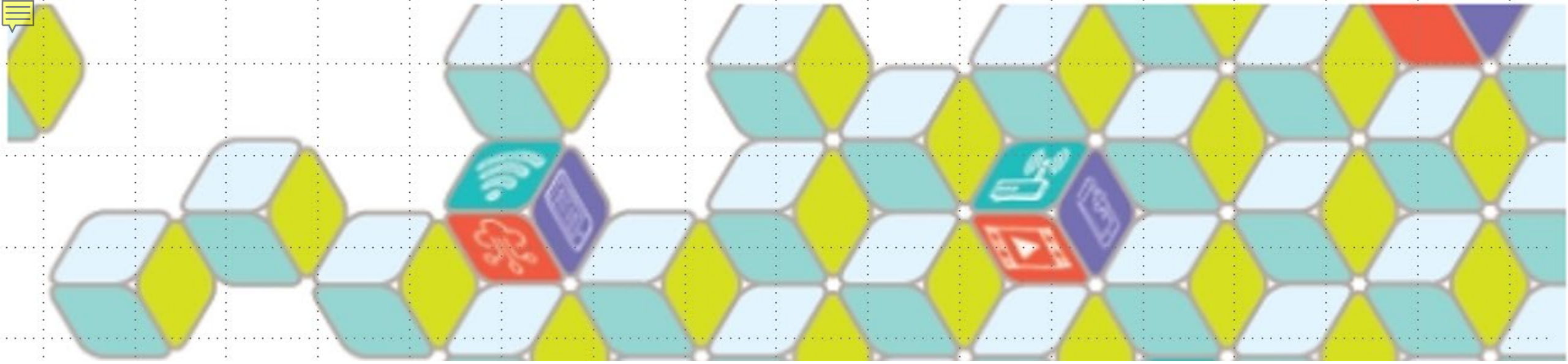


Programming Method
(using C, C++, for Games)

Prof. Tae Soo Yun
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DSU Dongseo University 동서대학교

Ch00_Introduction to Lecture_Unit01

International College

- International college : https://uni.dongseo.ac.kr/ic
- Dept.of Game (Korean Dept.) : https://uni.dongseo.ac.kr/game
  - Displayed students' activities, award, and graduation works
- Dept.of VideoAnimiation (Korean Dept.) : https://uni.dongseo.ac.kr/ani
  - Displayed students' activities, award, and graduation works

Introduction to the Lecture

❖Subject : Programming Method

❖Course Information

- C remains one of the most popular languages due to its portability and efficiency,
- Foundation for many programming languages like C++, Java, JavaScript, and Python.
- In this course, you'll learn the syntax of C(basic data types, control flow, function, array, file I/O), and how to properly apply them within an algorithm.
- Then, we'll create a games such as Tetris, Sokoban etc. based on console-game framework that provided by professor.

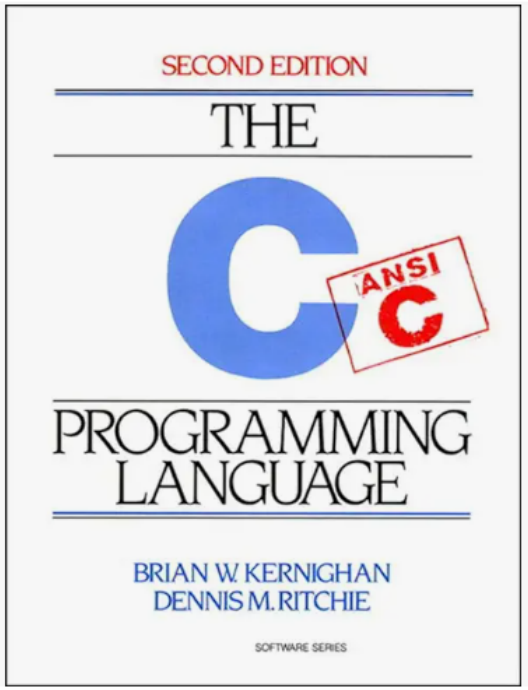
❖Teaching Methods

- Theory 1, Practice 2 hours.
- Provided exercise each class, should pass evaluation work(assignment).

Introduction to the Lecture (continue)

❖Textbook

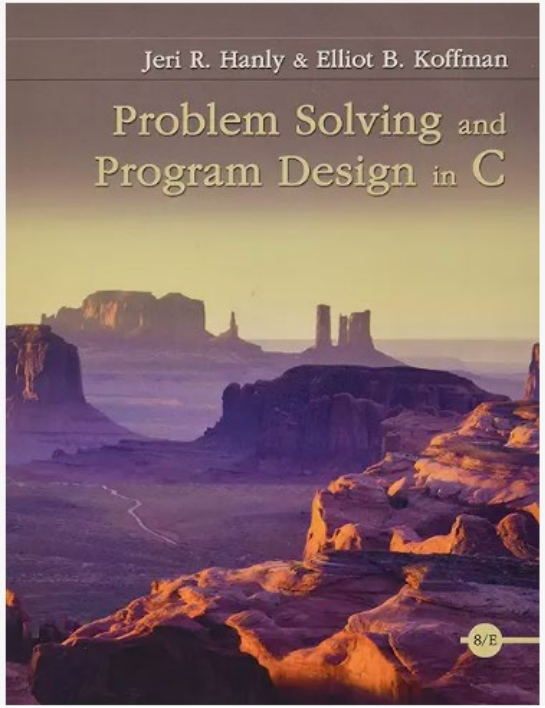
Main Textbook



C Programming Language, 2nd Edition

by Brian W. Kernighan and Dennis M. Ritchie

Supplementary Textbook



Problem Solving and Program Design in C

by Jeri Hanly and Elliot Koffman

\+

other materials from .....

You can download via https://url.kr/49k1ip

Introduction to the Lecture (continue)

❖ Participation

- Attendance: strongly encouraged except a personal death.
- Materials: Writing utensils, printed lecture materials, or a tablet (capable of note-taking); laptops are discouraged.
- The ability to summarize and understand lecture content through note-taking will be developed.
- I hope you will define your own measure of success and engage with the material in a way that best suits your needs.

❖ Academic Honesty

- All submitted work should be your own and academic dishonesty is not allowed. Academic dishonesty can be defined as :
  - ✓ Copying words, ideas, or other materials form another source without giving credit to the original author.
  - ✓ Copying answers or copying from your peers.
  - ✓ Empolying another person to alter or revise your work not referencing.
  - ✓ Using AI tools such as ChatGPT etc.

Syllabus

❖ The Contents to be studied

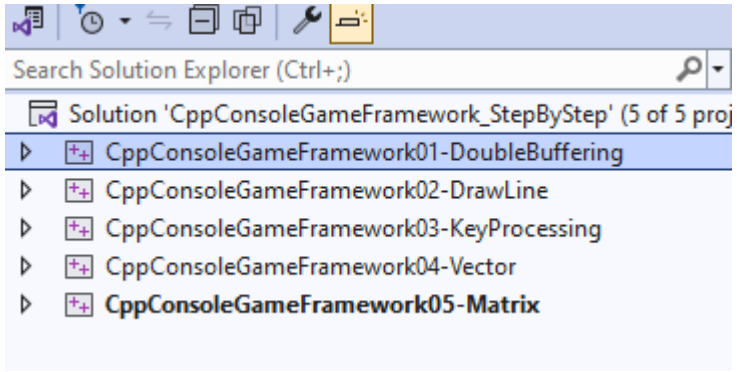
- Until midterm : C grammar, practical exercise
- After midterm : create a console based game using CbG & term project

week	Module	Contents	Etc.
1	(Module 1) Introduction	Course overview and Survey (learning Ability Assessment)	
2		C overview, Basic data types, Operators, and Expressions	Practice problems
3	Module 2	Control Flow	Practice problems, Assignments
4		Functions and Program Structure	Practice problems, Assignments
5		Pointers and Arrays	Practice problems, Assignments
6		Structures and Unions	Practice problems, Assignments
7		Input and Output Systems, File management	Practice problems, Assignments
8		Midterm examination	
9	Module 3	C++ basic grammar, OOP(Object Oriented Programming) Concept	Practice problems, Assignments
10	Module 4	Console based Game Framework	Practice problems, Assignments
11		Console based Game Framework (continue)	Practice problems, Assignments
12		Console based Game Framework (continue)	Practice problems, Assignments
13		Console based Game Framework (continue)	Practice problems, Assignments
14		Console based Game Framework (continue)	Practice problems, Assignments
15		Final examination & Presentation for term project	

Console based Game Framework

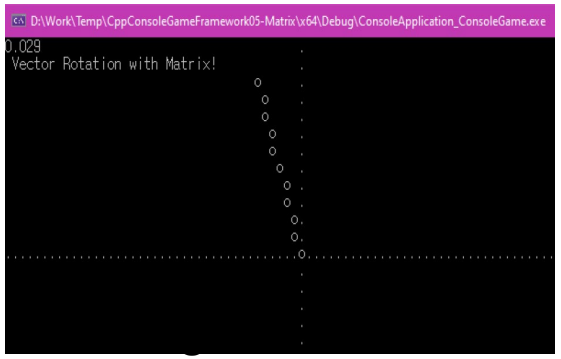
❖ Console based Game Framework

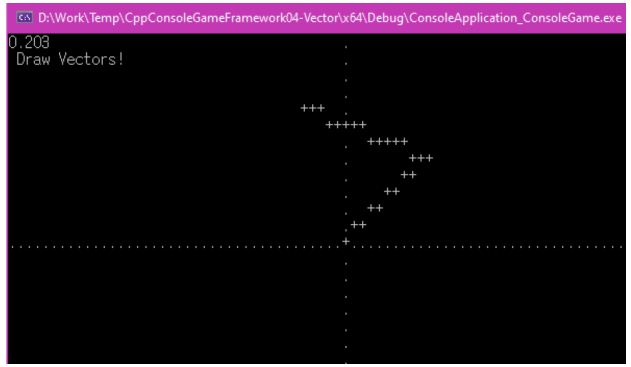
- DoubleBuffering : prevent n eliminate a blinkering for the smooth animation
- Drawline : bresenham line drawing algorithm is an efficient algorithm in computer graphics used primarily to draw a straight line connecting two points.
- Keyprocessing : code for processing several keyboard input and return the key value
- Vector,Matrix: key concepts for representing the movement of objects





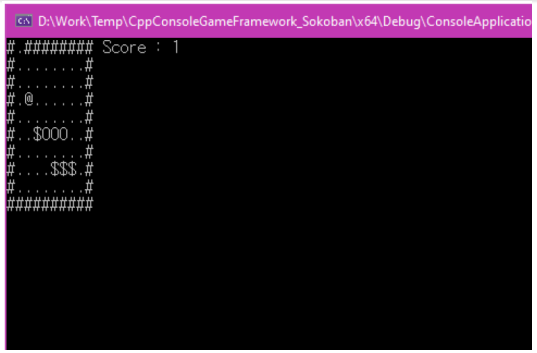


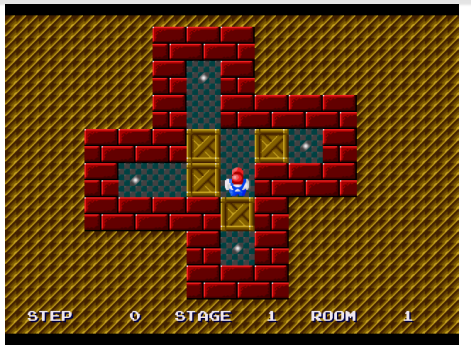




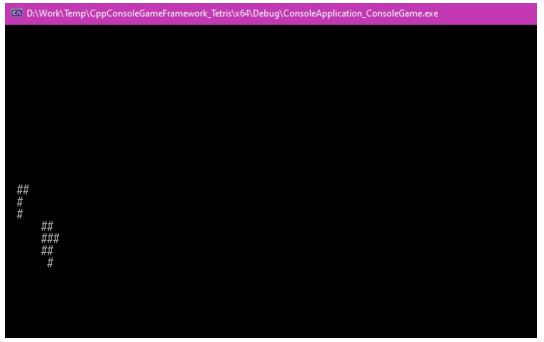
Console base GameFramework

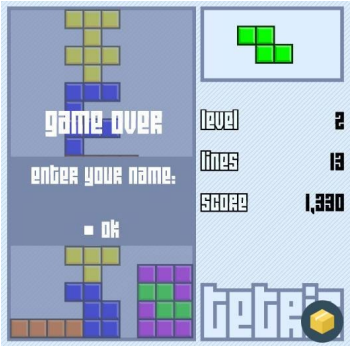
- Sokoban Games
  - The object of Sokoban is to take some objects (usuall boxes) to designated location by pushing them
  - How to play Sokoban





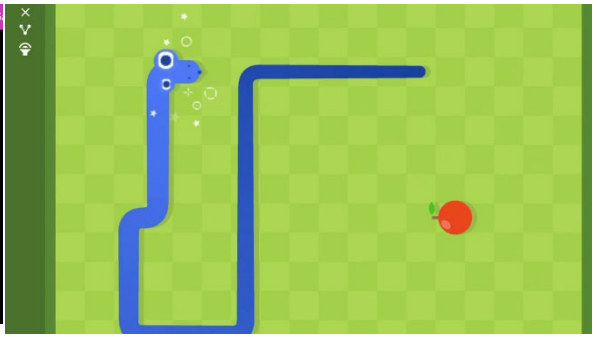
- Tetris Game
  - The goal of Tetris is to score as many points as possible by clearing horizontal lines of Blocks. The player must rotate, move, and drop the falling Tetrominos inside the Matrix (playing field)





- Snake Game
  - where you control a snake that grows longer by eating food items (apple)





Evaluation

❖ Evaluation Methods and Criteria

- Evaluation : Absolute Evaluation (flexible accord.to Univ. Policy)
- Midterm Exam : 20, Final Exam : 30, Attendance : 10, Assignment&Term project : 20, Learning Attitude : 10

❖Term project

- Write a Tetris game using console based game framework
- You have to write code to rotate the block and eliminate the line matched.(TBD)

평가방법 및 평가요소	절대평가									
	수시 평가	중간 고사	기말 고사	출석	과제물	토론	학습태도	발표	봉사 (튜더)	기타
		20	40	10	20		10			

Survey

- only used for preparing the class materials and teaching methods
- consists of 5 section
  - General Information
  - Programming background
  - C programming Expectations
  - Problem Solving Assesement
  - Etc.

- QR Code : https://forms.gle/jc92FJMp3wUNXaCw6



Q&A

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